ST. MARY'S DSG, KLOOF

PHYSICAL SCIENCES
Paper 2

GRADE 10

November 2024 **TIME: 2 hours + 10 min reading**
MARKS: 133

Examiner: T. Kirsten
Moderators: K. Bolton and G. Main

NAME: _____ TEACHER'S NAME:_____

READ THE FOLLOWING INSTRUCTIONS CAREFULLY:

1. This paper consists of a question paper of **15 pages**. Please check that your question paper is complete. A **data sheet** is also provided.

2. Read the questions carefully.

3. It is in your own interest to write legibly and to present your work neatly.

4. Answer all questions in the spaces provided.

5. **All working must be shown** and units may be used in substitutions, but **MUST** be given with answers. Work to 2 decimal places or significant figures where appropriate.

6. ONE blank page (15) is included at the end of the question paper. If you run out of space for a question, use this extra page. Clearly indicate the question number of your answer, should you use this extra space.

	Q1	Q2	Q3	Q4	Q5	Q6	Total	%
Mark								
Question total	**14**	**36**	**26**	**16**	**32**	**9**	**133**	**100**

QUESTION 1 – MULTIPLE CHOICE

Answer these questions on the multiple-choice answer grid. Make a cross (X) in the box corresponding to the letter that you consider to be the most correct.

1.1	A	B	C	D
1.2	A	B	C	D
1.3	A	B	C	D
1.4	A	B	C	D
1.5	A	B	C	D
1.6	A	B	C	D
1.7	A	B	C	D

1.1 An atom of a certain element has the following electron structure:

$1s^2 2s^2 2p^6 3s^2 3p^3$

The number of valence electrons, the valency and the charge of an ion of this element is:

	Valence é	**Valency**	**Charge of ion**
A	3	3	+3
B	5	5	+5
C	5	3	-3
D	3	5	-3

1.2 Which one of the following name and formula combinations is most correct?

	NAME	**FORMULA**
A	sulphur trioxide	SO_3
B	potassium dichromate	KCr_2O_7
C	zinc (I) hydroxide	ZnOH
D	ammonium nitrite	NH_4NO_3

1.3 Which one of the bonds between the atoms below is the *most ionic* in nature?

A H – C

B H – Cℓ

C H – O

D H – N

1.4 Which of the following formulae gives the correct number of molecules in 5 g of CH_4 (g)?

A $\frac{6,02\times10^{23}}{5\times16}$

B $\frac{5\times6,02\times10^{23}}{16}$

C $\frac{16\times6,02\times10^{23}}{5}$

D $\frac{5\times16}{6,02\times10^{23}}$

1.5 Which one of the following solids does NOT have a giant covalent network structure?

A Diamond

B Graphite

C Silicon dioxide

D Iodine

1.6 Which is the oxidising agent in the following reaction?

$$\mathbf{Br_2 + KI \rightarrow KBr + I_2}$$

A Br_2

B I^-

C Br^-

D I_2

1.7 The boiling point of CH_4 is much lower than that of HF. Which ONE of the following best explains this difference in boiling points?

A HF molecules are more polar than CH_4 molecules.

B CH_4 molecules are more polar than HF molecules.

C There are dipole-dipole forces between CH_4 molecules.

D There are hydrogen bonds between HF molecules.

7 x 2 = [14]

QUESTION 2

Refer to the following balanced equation, for a reaction taking place at STP:

$$\mathbf{4HNO_3(aq) + Pb(CO_3)_2(s) \rightarrow Pb(NO_3)_4(aq) + 2CO_2(g) + 2H_2O(\ell)}$$

2.1 Name the following compounds: (5)

a) $Pb(CO_3)_2$.

b) HNO_3

2.2 Describe an ionic bond. (3)

2.3 Which of the compounds in the reaction above are ionically bonded? (2)

2.4.1 Define an *electrolyte*. (2)

2.4.2 State one compound, in either the reactants or products in the reaction above, that could function as an electrolyte. Give a reason for your choice. (3)

2.5.1 Define *electronegativity.* (2)

2.5.2 With reference to electronegativity numbers, NAME and DESCRIBE the SPECIFIC intramolecular bond in CO_2. (4)

2.5.3 Name the intermolecular forces between CO_2 molecules. (1)

2.5.4 Depict a CO_2 molecule using Couper notation. (2)

2.5.5 How could one test that the gas released was CO_2? (2)

2.6.1 Show the formation of the bond in a water molecule, using Lewis diagrams. (3)

2.6.2 Name the intermolecular forces between water molecules. (1)

2.7 Calculate the volume of CO_2 produced when 235,44 g of $Pb(CO_3)_2$ reacts with excess HNO_3 at STP. (6)

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QUESTION 3

Refer to the following word equation:

silver nitrate + copper (II) sulphate → silver sulphate + copper (II) nitrate

3.1 Rewrite the equation in its balanced form, using formulae and phase symbols. (6)

3.2 What type of reaction is this? (1)

3.3.1 Calculate the molar mass of silver sulphate. (2)

3.3.2 How many moles is 112,32 g of silver sulphate? (2)

3.3.3 How many formula units is 112,32 g of silver sulphate? (3)

3.4 Consider the compound copper(II) sulphate. Describe and give the result of laboratory tests one could carry out to confirm:

a) the presence of the cation. (2)

b) the presence of the anion. (4)

3.5 Consider the following equation of an unbalanced chemical reaction:

$$Mn + H_3PO_4 \rightarrow Mn_3(PO_4)_2 + H_2$$

3.5.1 Balance the equation using suitable coefficients. (2)

3.5.2 Name the compound $Mn_3(PO_4)_2$ (3)

3.5.3 What type of reaction is this? (1)

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QUESTION 4

An iron nail was placed in a beaker containing a solution of blue copper (II) chloride.

- Changes were observed as in the diagrams below after 30 minutes.
- The blue colour of the solution started to fade.
- A red-brown deposit was seen forming on the iron nail.
- The beaker was warm to the touch.



Start



After 30 minutes

4.1 Complete the equation below, writing the products formed when iron reacts with copper (II) chloride. (2)

$$Fe + CuC\ell_2 \rightarrow$$

4.2.1 Define a *redox reaction*. (2)

4.2.2 Using oxidation numbers, show why the above reaction is a redox reaction. (4)

4.2.3 Identify the substance which was oxidised in the above reaction. (1)

4.3 Circle the correct term:

a) The above reaction is a COMBINATION/ DISPLACEMENT/ DECOMPOSITION reaction. (1)

b) The above reaction is an EXOTHERMIC / ENDOTHERMIC reaction. (1)

4.4 Give a reason that the blue colour stated to fade. (1)

4.5 Using an annotated diagram, name and describe the type of bonding in the iron nail. (4)

[16]

QUESTION 5

5.1 Sodium and potassium are both highly reactive metals with similar chemical properties, both found in group 1 of the Periodic Table

5.1.1 What name is given to Group 1 of the Periodic Table? (2)

5.1.2 Draw an Aufbau diagram of the potassium **ion**. (3)

5.1.3 Which other element has the same electron configuration as the potassium ion? (1)

5.1.4 Two stable (non-radioactive) isotopes of potassium exist in nature, potassium-39 and potassium-41. Potassium-39 comprises 93% of naturally occurring potassium, and potassium-41 accounts for the rest of naturally found potassium. Calculate the relative atomic mass of potassium. (3)

5.1.5 Write down the electron configuration of a sodium atom. (2)

5.1.6 Name the following compounds:

a) $NaCH_3COO$ (2)

b) $KCℓO_3$ (2)

5.1.7 Potassium is a more reactive metal than sodium. Explain this in terms of periodic trends. (4)

5.2 Refer to the following table showing the amount of energy required to remove successive electrons from two elements, X and Y, both found in **PERIOD 3**. Ionisation energies are shown up to the removal of the 8th electron in both cases (although both elements have more than 8 electrons in total)

Element	**Ionisation energies (kJ.mol^{-1})**							
	1st	**2nd**	**3rd**	**4th**	**5th**	**6th**	**7th**	**8th**
X	738	1451	7733	10 543	13 630	18 020	21 711	25 661
Y	1251	2298	3822	5159	6542	9362	11 018	33 604

5.2.1 Define *first ionisation energy*. (2)

5.2.2 What is the name of element X? Justify your answer with data from the table. (4)

5.2.3 Compare the 1st and 2nd ionisation energies of elements X and Y. Why are the ionisation energies of element Y that much greater? Use periodic trends to explain your answer. (4)

5.2.4 Name element Y. (1)

5.2.5 Write the formula of the compound formed when elements X and Y bond. (2)

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QUESTION 6

6.1 Describe an intermolecular force. (2)

6.2 Use only substances from the list below when answering Questions 6.2.1. to 6.2.7. The phase indicators (state symbols) represent the physical state of each of the substances at room temperature. The substances may be used more than once, or not at all. (7)

$N_2(g)$	**$NaC\ell(s)$**	**$SiO_2(s)$**	**$NH_3(g)$**
Ne(g)	**Fe(s)**	**CO(g)**	**graphite(s)**

Select one substance from the list that:

6.2.1 Has a giant covalent network structure between its identical atoms.

6.2.2 Has London forces between its atoms.

6.2.3 Has London forces between its molecules

6.2.4 Has dipole-dipole forces between its molecules.

6.2.5 Has hydrogen bonds between its molecules.

6.2.6 Has non-polar covalent bonds between its non-identical atoms.

6.2.7 Conducts electricity when melted into a liquid, but cannot conduct as a solid.

[9]

EXTRA SPACE